Fraido

Radical improvements in invasive access procedures.

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Our Solution*

*Patent Pending

A **fully mechanical** principle for **controllable diameter** in medical tube-shaped devices

The device's mechanical architecture enables controlled contraction and expansion, actuated directly by the clinician.

Key aspects of our diameter-regulation architecture are:

- **No plastic deformation**: diameter changes do not rely on permanent material deformation
- **Intrinsic actuation**: diameter change is generated by the tube's own structure, without additional expanding elements
- **Continuous**: any diameter within the operating range can be selected
- **Reversible**: the device can be repeatedly expanded and contracted

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Our Solution*

*Patent Pending

These characteristics translate into concrete clinical and technical **advantages**

No plastic deformation	Preserves material integrity over repeated diameter changes, improving mechanical reliability and reducing fatigue-related failure risks.
Intrinsic actuation	Eliminates the need for external expanding elements or auxiliary tools, simplifying device design and procedural workflow.
Continuous	Enables precise, real-time diameter modulation rather than discrete open/closed states, maximizing adaptability to anatomy and procedural demands.
Reversible	Allows repeated expansion and contraction at any point during the procedure, giving clinicians full control as clinical priorities evolve.

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Our Solution*

*Patent Pending

Our patent protects the application of this technology across multiple medical procedures.

Airways

- Endotracheal Intubation
- Lung selective Intubation
- Tracheostomy

Vascular

- Structural Heart
- Aneurysm Treatment
- Extracorporeal Oxygenation
- Pacing Devices

Drainage

- Abdominal Drainage
- Thoracic Drainage
- Urethral Catheter

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Our Solution*

*Patent Pending

Airways

- Endotracheal Intubation
- Lung selective Intubation
- Tracheostomy

Parallel development of all technologies was excluded:

- **Cost**: multiple certification, validation, and clinical evidence requirements
- **Technology validation**: core technology must be clinically and industrially validated in a first application before scaling
- **Organizational maturity**: demonstrated development ability, before expansion

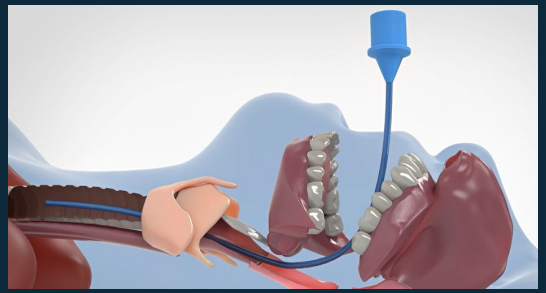
Endotracheal Intubation is an ideal first device:

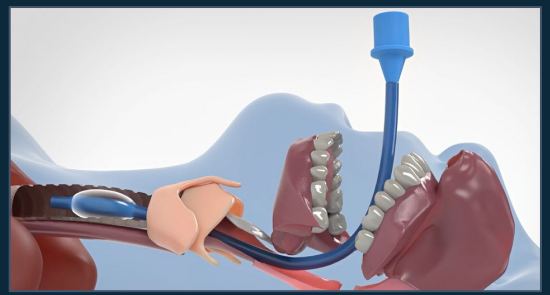
- **Lower technical risk**: larger dimensions reduce engineering complexity
- **Self-evident clinical benefit**: evident limitations of current devices, make our solution immediately attractive, facilitating adoption
- **Clear competitive benchmark**: effectiveness of the device is easily quantified using established benchmarks

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Standard of Care for Airway Management

When applied to Endotracheal Intubation procedures, our device is:





- Inserted thin
- Expanded once in place
- Adaptable during ventilation
- Collapsible for safer extubation

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Standard of Care for Airway Management

From mechanical features to clinical impact

Collapsible Diameter	Current devices must be inserted at full diameter; ours collapses for easier, safer entry.
Guided Insertion	Compatible with all currently available intubation-aiding devices.
Visibility	A slim profile at insertion to maintain full visibility throughout.
Adjustability	Real-time adjustability eliminates the need for tube exchanges, by adapting to changing clinical needs.

These features operationalize and extend the principles outlined in the Guidelines from the Difficult Airway Society[1].

[1]: DAS 2025 Intubation Guidelines — Difficult Airway Society, Br J Anaesthesia (2025)

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Standard of Care for Airway Management

	Current solutions (static diameter)	**Fraido** (controllable diameter)
Insertion	Tube diameter hinders insertion.	Insert small and expand in place.
Versatility	Tube size chosen according to patient's physiology.	One tube fits all.
Learning curve	Outcome depend on operator experience.	Simpler procedure for all personnel.

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Standard of Care for Airway Management

Difficulties during intubation have consequences for both patients and healthcare systems.

For **patients**, complications range from mild injury to life-threatening events, strongly correlated with prolonged time-to-intubation and repeated attempts.

For **hospitals**, additional costs arise from post-procedural treatment, complication management, and increased length of stay.

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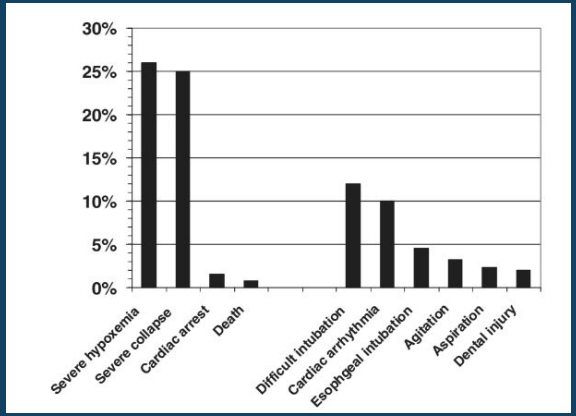
Standard of Care for Airway Management

For patients, the key determinants of outcome are the number of attempts and time to effective ventilation.

Likelihood of complications increases as the number of **failed attempts** grows[1].

Attempt	Likelihood of complications
2	x 4.4
3	x 7.4
4	x 13.9

Intubation-related complications span a wide range of clinical severity[2].



[1]: April MD, Schauer SG, Nikolla DA, Casey JD, Semler MW, Ginde AA, Carlson JN, Long BJ, Brown CA 3rd. Association between multiple intubation attempts and complications during emergency department airway management: A national emergency airway registry study. Am J Emerg Med. 2024 Nov;85:202-207. doi: 10.1016/j.ajem.2024.09.014. Epub 2024 Sep 11. PMID: 39288499.
[2]: Jaber S, Amraoui J, Lefrant JY, Arich C, Cohendy R, Landreau L, Calvet Y, Capdevila X, Mahamat A, Eledjam JJ. Clinical practice and risk factors for immediate complications of endotracheal intubation in the intensive care unit: a prospective, multiple-center study.Crit Care Med. 2006 Sep;34(9):2355-61. doi: 10.1097/01.CCM.0000233879.58720.87. PMID: 16850003.

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Standard of Care for Airway Management

The risk of intubation-related complications increases significantly in specific patient populations.

Risk-Increasing Factors	Major Complication Probability [odds ratio] [1]
Obesity	x 2.2
Tumors	x 1.9
Cardiovascular Disease	x 1.4
Diabetes	x 1.1

[1]: Moucharite MA, Zhang J, Giffin R. Factors and Economic Outcomes Associated with Documented Difficult Intubation in the United States. Clinicoecon Outcomes Res. 2021 Apr 1;13:227-239. doi: 10.2147/CEOR.S304037. PMID: 33833535; PMCID: PMC8021135.

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Standard of Care for Airway Management

The clinical consequences of difficult intubation translate into **higher costs** and **longer hospital stays**.

Benchmark parameter	Easy Intubation	Difficult Intubation
Total Cost of Permanence	$ 12k	$ 33k
ICU Length of stay	3.4 days	5.4 days
ICU admission post OR	16 %	60 %
Cost of ICU	$7k - $9k	$12k - $16k

[1]: Yoon, U., Mojica, J., Wiltshire, M. et al. Emergent airway management outside of the operating room – a retrospective review of patient characteristics, complications and ICU stay. BMC Anesthesiol 19, 220 (2019). https://doi.org/10.1186/s12871-019-0894-4
[2]: Moucharite MA, Zhang J, Giffin R. Factors and Economic Outcomes Associated with Documented Difficult Intubation in the United States. Clinicoecon Outcomes Res. 2021 Apr 1;13:227-239. doi: 10.2147/CEOR.S304037. PMID: 33833535; PMCID: PMC8021135.

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Standard of Care for Airway Management

U.S. data highlight the clinical and economic scale of failed first-attempt intubations in emergency settings[1].

In emergency settings in the US, the average **First Pass Success Rate** is estimated around **84%**.

Managing the consequences of the remaining 16% of failed first attempts results in an estimated **annual** cost of **USD 890 million**.

[1]: Estimating the Economic and Absolute Number of Complications Associated with Emergency Intubations Performed Outside the Operating Room; A Methodology for Estimating the Burden in the United States.

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Market & Business Model

Our market opportunity is best framed by the Endotracheal Tube market.

GLOBAL ENDOTRACHEAL TUBE MARKET

Market size	Market growth
$3.1B	**CAGR 6.3%**

To accurately capture market potential and meaningful segmentation, a bottom-up analysis is required.

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Market & Business Model

U.S. intubation volumes were used as a reference due to reliable and cross-verifiable data.

Using the U.S. as a reference, annual intubation volumes were extrapolated for additional global markets, adjusting for population size and national healthcare expenditure.

Additional market segmentation was performed based on care setting and procedure type.

Intubations segmented by care setting reflect distinct clinical contexts and workflows:

- Operating Room (OR)
- Intensive Care Unit (ICU)
- Emergency Department and pre-hospital emergencies (ED)

Intubations segmented by procedural type reflect different levels of urgency and complexity:

- Emergency intubation
- Difficult intubation
- Pediatric intubation
- Nominal intubation

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Market & Business Model

We initially target **non-nominal intubations** (emergency and difficult), where unmet clinical needs are most acute and the value of our solution is immediately apparent, supporting rapid adoption and real-world validation.

Following this entry phase, we will expand into **nominal intubations** to capture the largest volume segment of the market.

Preliminary **device pricing** has been defined based on:

- Benchmark pricing of existing devices
- Total material cost of standard intubation procedures[1]
- Published pricing references for innovation in airway management[2]

Revenue predictions conservatively neglect:

- Sector growth (population growth, emerging markets, CAGR)
- Increase in comorbidities (obesity, age,...)
- Increasing variability in operator experience

[1]: Moucharite MA, Zhang J, Giffin R. Factors and Economic Outcomes Associated with Documented Difficult Intubation in the United States. Clinicoecon Outcomes Res. 2021 Apr 1;13:227-239. doi: 10.2147/CEOR.S304037. PMID: 33833535; PMCID: PMC8021135.
[2]: Maguire S, Schmitt PR, Sternlicht E, Kofron CM. Endotracheal Intubation of Difficult Airways in Emergency Settings: A Guide for Innovators. Med Devices (Auckl). 2023 Jul 18;16:183-199. doi: 10.2147/MDER.S419715. PMID: 37483393; PMCID: PMC10362894.

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Market & Business Model

Combining our projected penetration with the number of intubations, evaluated according to our model, it is possible to calculate expected sales and revenues.

Projected **market penetration**:

	Region	Year 1	Year 2	Year 3	Year 4	Year 5
Non-Nominal	North America Europe	0.0 % 0.5 %	0.0 % 2.5 %	1.0 % 6.5 %	3.0 % 13.0 %	9.0 % 18.0 %
Nominal	North America Europe	0.0 % 0.05 %	0.0 % 0.1 %	0.05 % 0.5 %	0.1 % 1.5 %	0.5 % 3.0 %

Projected **sales**:

Region	Year 1	Year 2	Year 3	Year 4	Year 5
North America Europe	€ 0 € 590.000	€ 0 € 2.200.000	€ 1.010.000 € 7.020.000	€ 2.780.000 € 16.400.000	€ 9.300.000 € 27.100.000
Total	€ 590.000	€ 2.200.000	€ 8.030.000	€ 19.200.000	€ 36.400.000

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Market & Business Model

Sales require:

- Reliance on a distributor
- Established **reimbursement pathway**

Europe

Nominal: **18.990.000** Non-Nominal: **2.860.000**

Country	Intubation market share	Cumulative market share
Germany	17.96 %	17.96 %
UK	13.46 %	31.42 %
France	12.99 %	44.41 %
Italy	9.04 %	53.45 %
Spain	7.36 %	60.82 %
Others (23)	39.18 %	100 %

North America

Nominal: **20.120.000** Non-Nominal: **3.030.000**

Country	Nominal Intubations	Non-Nominal Intubations
USA	18.536.327	2.791.699
Canada	1.584.267	238.602

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Market & Business Model

Europe

Nominal: **18.990.000** Non-Nominal: **2.860.000**

Country	Intubation market share	Cumulative market share
Germany	17.96 %	17.96 %
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France	12.99 %	44.41 %
Italy	9.04 %	53.45 %
Spain	7.36 %	60.82 %
Others (23)	39.18 %	100 %

- Single regulatory approval (MDR, UK exception)
- Established clinical connections

- Fragmented reimbursement pathways

North America

Nominal: **20.120.000** Non-Nominal: **3.030.000**

Country	Nominal Intubations	Non-Nominal Intubations
USA	18.536.327	2.791.699
Canada	1.584.267	238.602

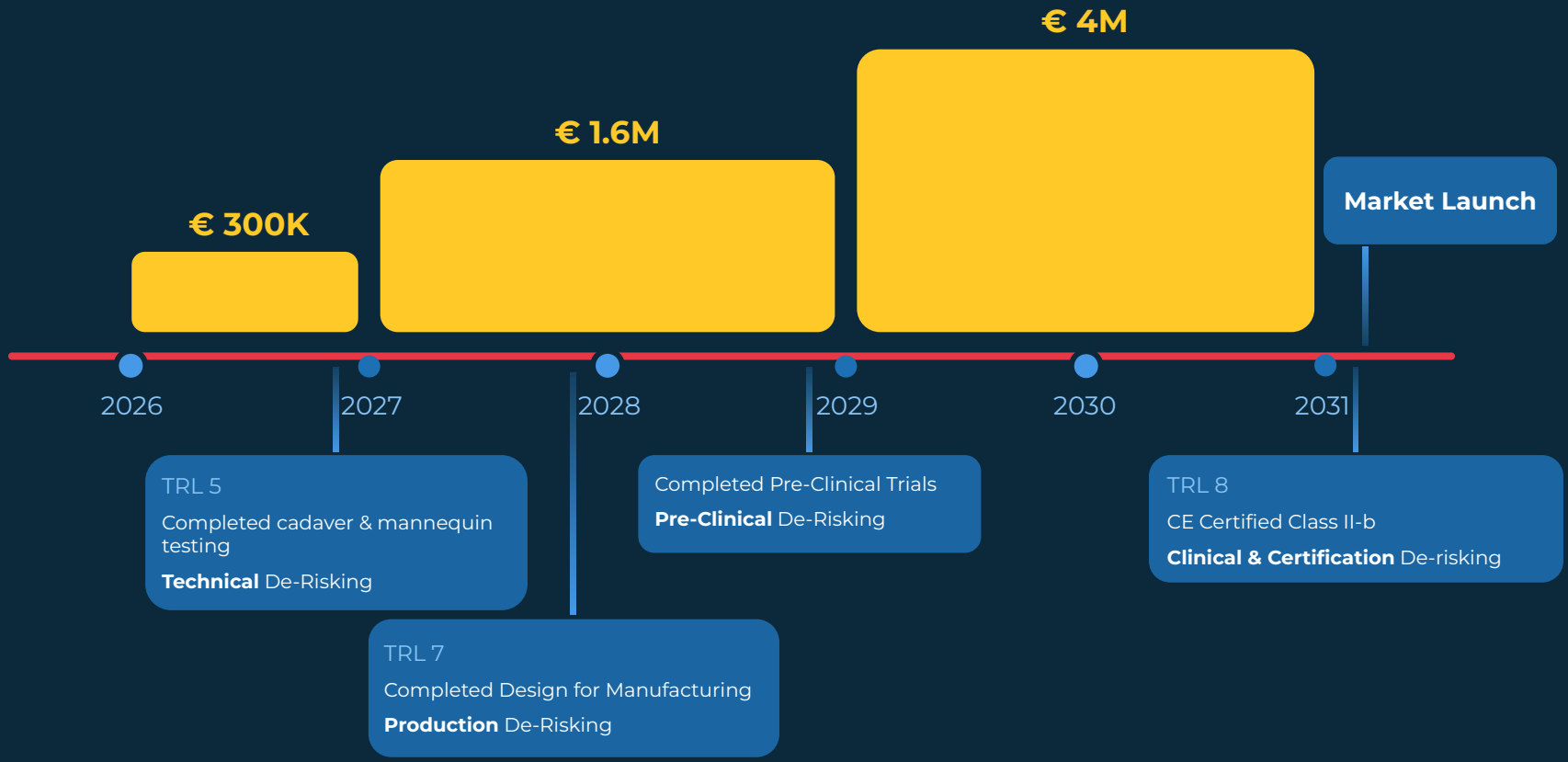
- Two regulatory approvals (FDA + MDL)
- Two reimbursement strategies

- Weaker clinical connections to date

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Timeline: Funding & Regulatory



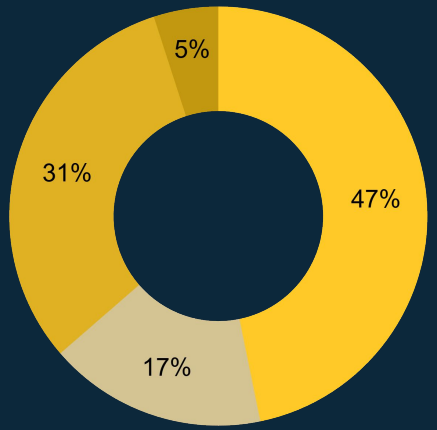
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Timeline: Funding & Regulatory

€ 300K	€ 1.6M	€ 4M

2026 2027 2028 2029 2030 2031



R&D / Clinical	47%
Wages & Rents	31%
Legal	17%
Patent Extension	5%

Raised via SAFE agreement
- € 60k already committed
- € 50k raised via non-dilutive opportunities

Complete functional prototyping of the device.

Early clinical validation through structured usability and performance assessment.
In-vitro trials on mannequins and cadavers to benchmark performance parameters.

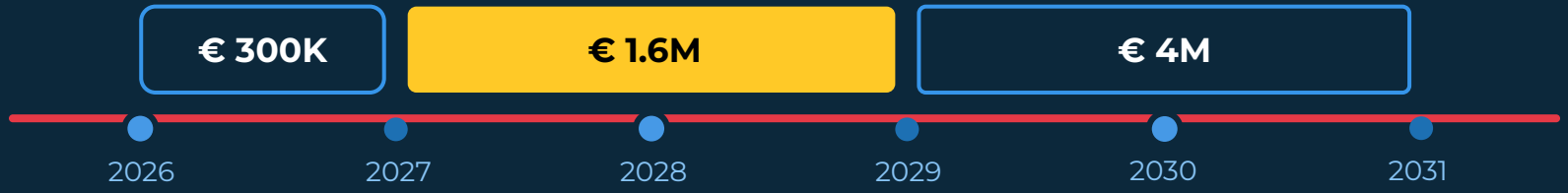
Draft regulatory technical documentation to support MDR compliance.

International IP extension through PCT filing.

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Timeline: Funding & Regulatory



Complete Design for Manufacturing (DFM):

- Device design freeze
- Consolidated manufacturing process

Completed pre-clinical regulatory requirements:

- Biological testing
- Mechanical testing

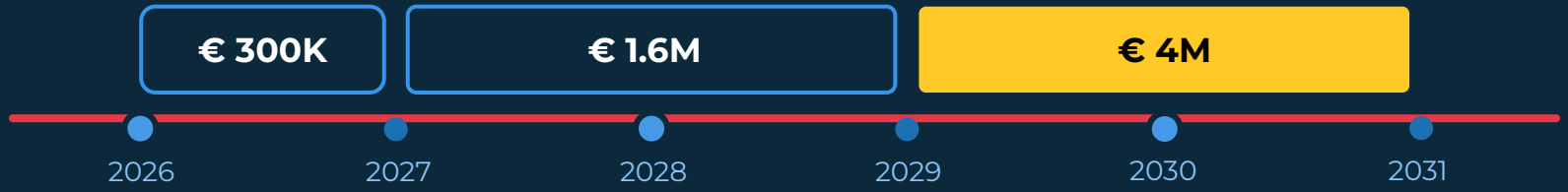
Note: The current budget conservatively covers full biological and mechanical testing, although a reduced test scope may ultimately be sufficient.

Set-up of company's QMS

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Timeline: Funding & Regulatory



Completion of regulatory validation:

- Clinical Trials
- Submission for MDR conformity assessment

Finalization of market agreements:

- Full scale manufacturing agreement
- Finalized distribution agreement

Market launch and GTM activities

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Our Solution - Patent Timeline

Currently supported by Studio Torta

04/2025 Internal SoA Research	Internal state of the art research on existing patents. 27 other patents considered (airways and vascular). Identification of primary features and technologies.
07/2025 Italian Patent Deposition	Deposition of Italian patent with the support of Studio Torta.
01/2026 FtO Return	Return of official Freedom to Operate research. 4 overlapping patents identified. Non-problematic overlapping linked to general wording of claims, to be addressed during PCT extension.

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Our Solution - Patent Timeline

Currently supported by Studio Torta

07/2026 PCT Extension	International extension of patent via Patent Cooperation Treaty. Implementation of Italian FtO return. Cost covered by 3i Voucher (already confirmed).
04/2027 Italian Patent Review	Deadline for the review of the Italian patent based on FtO return.
01/2028 International Pat. Ext.	Deadline for the extension of the patent to other countries. N.B.: priority linked to Italian patent deposition (07/2025).

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Fraido’s Team

Antonio Maria Vizioli
CEO & Co-Founder

Nurse with 5 years of experience in emergency care.

User-driven understanding of **clinical** needs.
Leverages a strong clinical **network** to guide development and validation.

Full-time on the project.



Elia Fregonese
CTO & Co-Founder

Materials and nanotechnology **engineer** with 4 years of experience in the space sector.

Experienced in long **development** cycles and high-reliability systems.
Relevant experience in **ISO-compliant** developments.

Full-time on the project.

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Fraido's Network

Advisors

Prof. Federico Piccioni - Head of ICU, Humanitas Hospital

Prof. Pietro Arina - Anesthetist, Professor UCL London

Dr. Chiara Sansovini -Anesthetist , Autor Stargate Study

Dr. Roberto Righetti - **Marco Garroni -** Airway Instructors

Prof. Giacomo Bellani - Head of ICU, Trento Hospital

Dr. Stefano Bonvini - Head of Vascular Surgery

Dr. Andrew Markle - Anesthetist, Saint Mary London

Martin Gossling - Clinical Engineer UHS, NHS Mentor

Hospital Network

Italy

Meyer Hospital Florence - Careggi Hospital Florence - Humanitas Hospital Milan - San Raffaele Hospital Milan - Padova Hospital - Ravenna Hospital - Bambino Gesù Hospital Rome - Perugia Hospital - Torino Hospital - S.Chiara Hospital Trento - Borgo Roma Hospital Verona

HEMS Aosta - HEMS Perugia

UK

Liverpool Hospital - Southampton Hospital - Saint Mary's Hospital London

N.B.: Anesthetists from the hospitals listed above have signed LOIs.

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Let's Transform Invasive Access

Together,
we can redefine safety and simplicity in invasive access.

Curious? Have a look.

www.fraido.it

Questions? Let's talk.

info@fraido.it

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